Waste Sorting Image Classification

6-class waste classifier (cardboard, glass, metal, paper, plastic, trash) on TrashNet, comparing two ImageNet-pretrained CNN backbones (EfficientNet-B0, ResNet-50) and two fine-tuning strategies (feature extraction vs. full fine-tuning).

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Layout

```
configs/                one YAML per experiment (baseline:
baseline_effb0_fe.yaml)
scripts/prepare_data.py  hash, deduplicate and split the dataset ->
data/splits.csv
src/data.py              Dataset, transforms, stratified split, class
weights
src/models.py            timm backbone + new head, freezing, LR groups
src/train.py             training loop (AdamW, warm-up + cosine, early
stopping on val macro-F1)
src/evaluate.py          test metrics, confusion matrix, misclassified
list, latency benchmark
runs/<name>_seed<k>/     checkpoints and results (git-ignored except small
json/png)
```

Quick start on Linux or MacOS

```
pip install -r requirements.txt

# 1. data (~43 MB)
mkdir -p data/raw && cd data/raw
curl -L -o ds.zip
https://raw.githubusercontent.com/garythung/trashnet/master/data/dataset-
resized.zip
unzip -q ds.zip && rm -rf __MACOSX ds.zip && cd ../..
python scripts/prepare_data.py --root data/raw/dataset-resized

# 2. ImageNet weights (or set weights_file: null to let timm download them)
mkdir -p weights
curl -L -o weights/efficientnet_b0_ra-3dd342df.pth
https://github.com/rwightman/pytorch-image-models/releases/download/v0.1-
weights/efficientnet_b0_ra-3dd342df.pth

# 3. baseline
python -m src.train    --config configs/baseline_effb0_fe.yaml
python -m src.evaluate --config configs/baseline_effb0_fe.yaml
```

Alternative quick start

If you have **pixi** installed on your system,

1. Downloand and unzip dataset:

```
curl -o ds.zip
https://raw.githubusercontent.com/garythung/trashnet/master/data/dataset-
resized.zip
```

Create necessary folders and place it into ./data/raw/dataset-resized

2. Download weights and place them into ./weights folder

```
curl -o weights/efficientnet_b0_ra-3dd342df.pth
https://github.com/rwightman/pytorch-image-models/releases/download/v0.1-
weights/efficientnet_b0_ra-3dd342df.pth
```

3. Install dependencies, prepare data and run the baseline:

```
pixi install
pixi run prepare
pixi run train
pixi run evaluate
```

Status

- [x] Data pipeline (deduplication, stratified 70/15/15 split, augmentation)
- [x] Baseline: EfficientNet-B0, frozen backbone (E-FE) — test accuracy 84.7%, macro-F1 0.824
- [ ] E-FT, R-FE, R-FT configurations, 3 seeds each
- [ ] Grad-CAM failure-case analysis